

RUTURAJ MARATHE

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SUMMARY

Backend Software Engineer with 3+ years of experience building automation-heavy backend services, data-processing pipelines, and reliable distributed systems in production environments.

PROFESSIONAL EXPERIENCE

Visteon Corporation

Software Developer – AI & Innovation, IT, Automation

Van Buren, MI

Jun 2025 – Present

- Designed and deployed FastAPI-based microservices to automatically generate complex automotive test reports, eliminating 5,000 manual hours per operator annually.
- Developed RAG pipeline to extract and index unstructured engineering artifacts, enabling conversational document search and improving data access efficiency by 70%.
- Implemented autonomous Bill of Materials costing, reducing hardware architecture turnaround time by 50%.
- Engineered PL/SQL delta-load procedures to migrate and process 5M+ records across 7+ reports, improving query performance and data governance.

Intel Corporation

Software Engineer Intern – Backend / ML Systems

Santa Clara, CA

May 2024 – Dec 2024

- Designed REST API-driven automation to programmatically create and track defects, cutting bug reporting time by 70% for 50+ silicon design engineers.
- Developed scalable data-processing pipelines to analyze 100K+ timing paths, automatically surfacing Setup/Hold violations 2 design iterations earlier and reducing late-stage failures.
- Implemented ML based predictive backend logic achieving 85% accuracy, reducing I/O budgeting iterations from 10 to 6 cycles.
- Collaborated cross-functionally to build multi-select defect logging feature with database state management, improving productivity for 50+ global engineers filing HSD bugs.

Visteon Technical and Services Center

Software Engineer – Automotive Systems

Bangalore, India

Jan 2021 – Jul 2023

- Implemented backend service APIs in C/C++ supporting Java API-layer calls in an Android Automotive system, deployed across 500+ production infotainment units.
- Designed persistent CRUD workflows on media databases enabling favorites bookmarking and fast-access playback, reducing user navigation steps by 60%.
- Built a fault-tolerant factory reset and recovery pipeline preserving media device state and achieving 100% recovery reliability across device reboot.
- Debugged and resolved 45 production defects across Java APIs, native middleware, and Linux system calls in a multi-threaded environment.

TECHNICAL SKILLS

Programming & Data	Python, Java, C, C++, SQL, Data Modeling, Query Optimization
Frameworks & Libraries	REST APIs, FastAPI, Django, Selenium, Postman, Scikit-learn, Requests
Cloud & DevOps	AWS (EC2, S3, Lambda), Azure(OpenAI, Cost monitor) Docker, CI/CD, Git
Systems Engineering	Linux, Multithreading, IPC, Cross-Compilation, Android Automotive (AOSP), Cross-layer Debugging (Java, Native, Linux)

PROJECTS

DoctorAssist.AI — VectorDB, RAG, NLP, Gemini API

- Built a backend service enabling natural-language querying over relational healthcare data, translating user queries into optimized SQL and improving data access speed by 40%.

Used Car Data Analysis (AWS, EC2, S3, boto3, PySpark)

Nov 2024

- Pioneered a scalable pipeline to extract pricing insights from used car listings, improving efficiency by 25%.

EDUCATION

Indiana University Bloomington

M.S. in Computer Science - GPA: 3.87 - Software Engineering, Applied Machine Learning

Bloomington, IN

May 2025

Visvesvaraya Technological University

B.E. in Computer Science - GPA: 8.08 - Data structures, Object Oriented Programming

Belgaum, India

Aug 2020